Ringbot Fabrication Final Report

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ME 497: KIMLAB Independent Study

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1. Introduction

Ringbot is a monocycle robot with leg modules that allow it to drive forward, balance, and stand itself up. Much of the design has already been completed by prior contributors; my main goal is to fabricate an entire Ringbot for KIMLAB's future use, making improvements to the design along the way. My goal was also to learn more about the design of robotic systems and to understand why certain design choices were made with respect to mechanical and electrical aspect of the project, and see if I could improve upon some of these choices.

2. Project Timeline

Timeframe	Key Milestones
Week 1	Onboarding
Week 2	Working on outer rim
Weeks 3-6	Tire iteration and fabrication
Weeks 7-8	Primary and secondary module fabrication and assembly
Weeks 9-12	Wiring and electronics focusing on primary module
Week 13	Leg fabrication and assembly
Weeks 14-16	More electronics, finishing up loose ends, final pres and report

Table 1: Project timeline

3. Team Roles

I worked on this solo. Responsibilities encompassed the entire project lifecycle, from component fabrication and assembly to wiring, electronics, and configuring the simulation environment.

4. Methods & Results

4.1 TPU Tire Fabrication

Initial attempts to print the entire tire using solely TPU resulted in dense structures with difficult-to-remove supports. To address this, the dual-extrusion capability of the Raise3D E2 printer was utilized to print PLA supports for the TPU main body. Slicing was performed using ideaMaker, building upon baseline project templates. Early print failures at a 55°C bed temperature were resolved by increasing the temperature to 60°C and, more importantly, performing a precise z-calibration on the right extruder to ensure proper adhesion. While the initial dual-material print succeeded, a slicing issue occurred where an unintended interfacial layer of TPU was generated between the main body and the PLA supports as shown in Figure 1. This was adding an extra layer of TPU that was very difficult to remove from the main body.

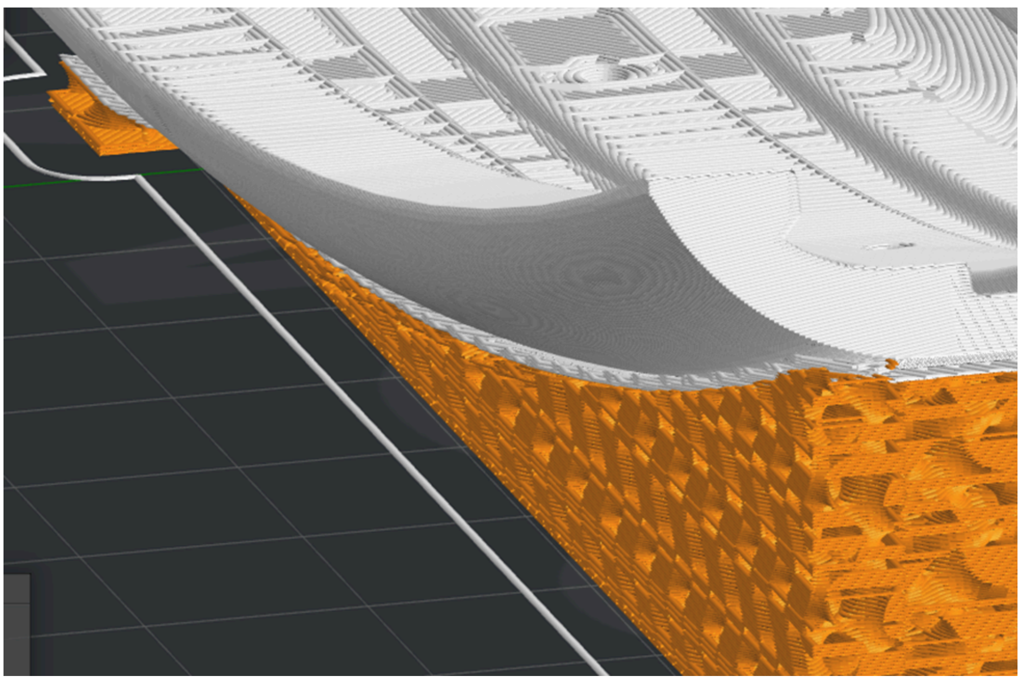

Figure 1: Slicer adding extra layer of TPU

This turned out to be a simple fix of change the dense support material from TPU to PLA. However, it was still difficult to remove the PLA supports from the TPU body. I then adjusted a large number of settings to make the PLA supports as brittle as possible in an attempt to make them easier to break off. This resulted in a support removal time of roughly 20 minutes, which is long, but much shorter than the initial times of nearly an hour. Figure 2 shows the before and after of all of these setting changes.

Figure 2: Initial vs. final dual extruded tire prints

4.2 Drive Module Assembly

The ringbot has two drive modules, a primary module with a Raspberry Pi mounted, and a secondary module without the Pi. The assembly of each drive module required two servos, the drive module body, the drive module cover, and a specific selection of fasteners and custom gears. To prevent mechanical interference with the servo housing during operation, shorter M1.5 screws were utilized for the flange mounting.

The spur gear assembly required several dimensional iterations to achieve proper functional tolerances. The nominal diameter for the bearing and washer is 16mm, while the initial CAD dimension for the housing hole was 16.4mm (measuring 16.35mm on the physical print). The CAD model was adjusted to a precise 16mm diameter. While this allowed the bearing to seat securely, the resulting gear profile was too thick. To address this, the gear thickness was reduced from 6.4mm to 5.8mm. This modification initially yielded successful gear meshing when mounted to the rack; however, the assembly became overly tight, likely due to excessive friction from the thrust bearings. Increasing the recess hole depth by 0.3mm successfully mitigated this binding issue. Additionally, the central shaft hole diameter was insufficient for the M3 screws to thread properly. The hole was manually drilled out, increasing the diameter from 2.6mm to 2.8mm, a modification that will be updated in the digital models for future iterations.

During testing, it was observed that the fasteners securing the flange to the gear protruded excessively, causing frictional rubbing against the servo during rotation. Replacing these with shorter screws resolved the immediate interference; however, it introduced a secondary friction point where the gear face rubbed directly against the drive module chassis. Measurements indicated a 3.9mm distance from the servo to the gear surface, compared to a 3.75mm thickness of the drive module body. To provide adequate clearance, the mounting face of the gear was extruded by an additional 0.3mm in the CAD software. This adjustment successfully eliminated all frictional contact between the gear and the module body while maintaining meshing between the large and small gears. From this point, both drive module were assembled smoothly.

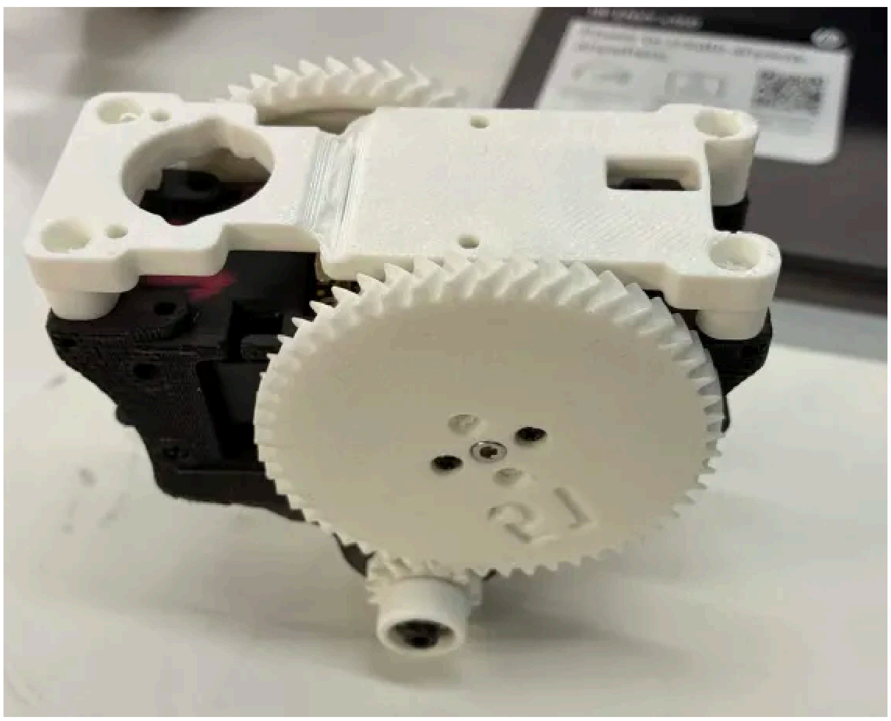

Figure 3: Drive module assembly before wiring and electronics

4.3 Electronics and Wiring Integration

The electrical integration of the primary drive module required establishing reliable power and data connections between the controllers (U2D2 and Raspberry Pi), the power distribution system (voltage regulator, batteries), and the IMU. Additionally, robust inter-module communication lines had to be fabricated and routed to connect the primary and secondary drive modules.

Initial assembly involved iterative soldering and routing adjustments to successfully interface the Raspberry Pi with the voltage regulator, the U2D2 motor controller, and the IMU.

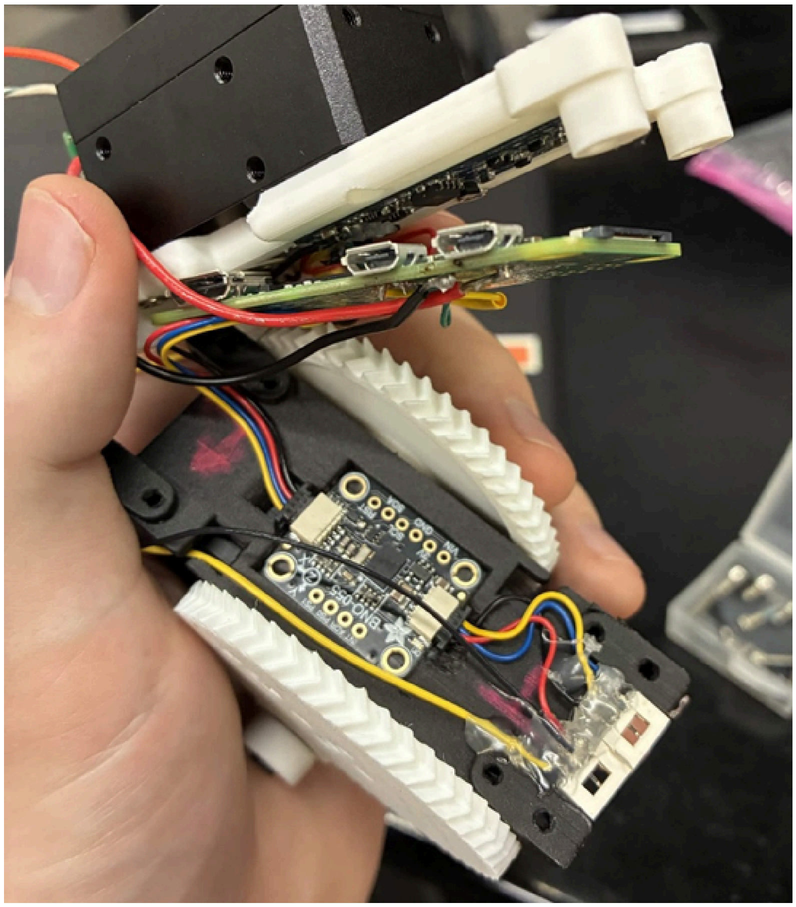

Figure 4: Partially complete drive module assembly

To facilitate communication between the modules, custom wiring harnesses were fabricated using 6-conductor telephone wire. This required crimping to properly interface the wires with male connectors. The female connectors were then soldered to their corresponding signal lines within the drive module and secured in place using hot glue.

Following the installation of the inter-module connectors, a mechanical clearance issue was identified during the final assembly of the module chassis. The glue profile securing the connectors was too high, preventing the Raspberry Pi and the protective cover from seating correctly. The excess glue was removed to avoid damaging the fragile solder joints on the underlying female connectors. The components were then re-secured, and the glue was compressed using the broad face of wire cutters to achieve a flat, low profile. This structural modification provided the necessary clearance, allowing for the successful final assembly of the Raspberry Pi and the module cover.

4.4 Perfboard and Power Distribution

Next I began working on the perfboard that connects all of the motors to the batteries, each other, and the U2D2. At first I followed the previous designs, using lines of solder on the bottom of the perfboard to connect all power, ground, and data ports of the connectors to each other. This was a very volatile system, as one little mistake would cause these solder lines to fuse as seen above, and this was very difficult to undo.

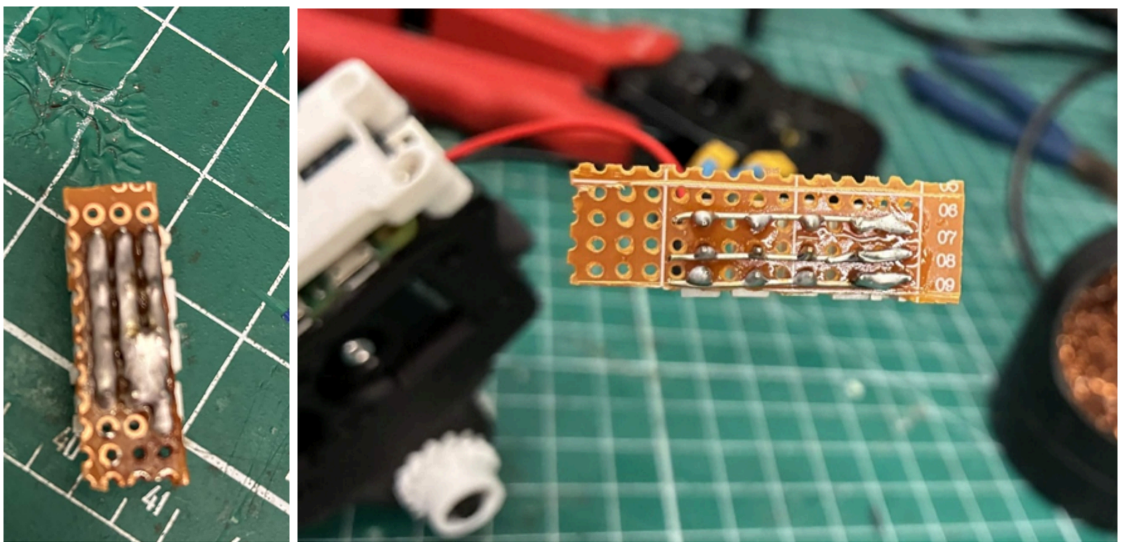

Figure 5: Left shows fused solder lines, right shows solution using solid core wire

I then decided to use a piece of stripped solid core wire to connect the connectors to each other, simply soldering the connectors to the wire instead of using lines of solder. This worked quite well and I think should be used in future iterations. Figure 6 shows the battery connector, which was made by making two 3-way splices of wire.

Figure 6: Battery connector

The end result connects two batteries and the voltage regulator to the perfboard, ensuring the voltage regulator wires route underneath the body cover and out to the other side of the drive module to attach to a connector on the voltage regulator.

4.5 Leg Assembly

The motors mounted on the legs utilize the HN12-I101 motor horn, which allows for wires to route through the motor and out of the back, directly opposite the active shaft. This prevents wire tangling and helps with wire harnessing. After attaching the new horns and routing the wires through the motor, I began assembling the leg using parts I had 3D printed with the P1S. I quickly ran into an issue with screw interference. The smallest screw size I could find for M2 screws in the lab, 5mm, was too long, and would poke through the standard servo horn and scratch the motor surface, as shown below.

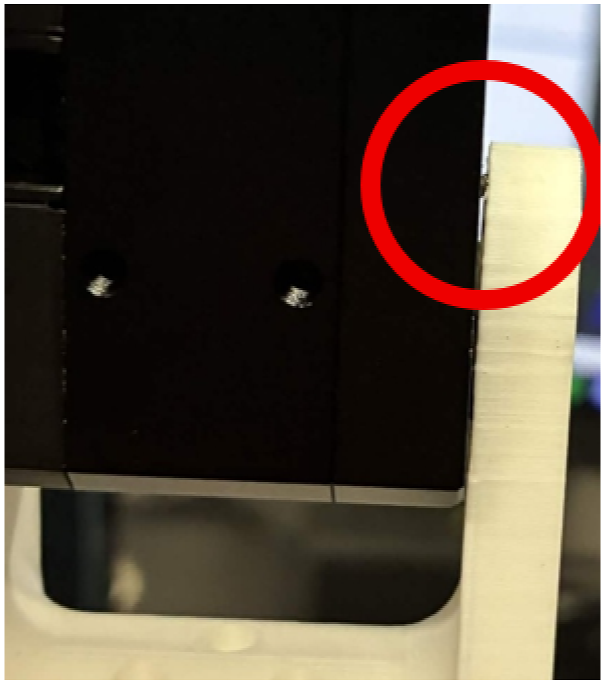

Figure 7: Screw interference causing scratch on motor surface

This caused heavy friction, so I decreased the depth of the counter bore on the white part pictured below by 0.5mm and reprinted the part. This fixed the issue. The rest of the assembly went quite smooth, yielding the following two Ringbot legs.

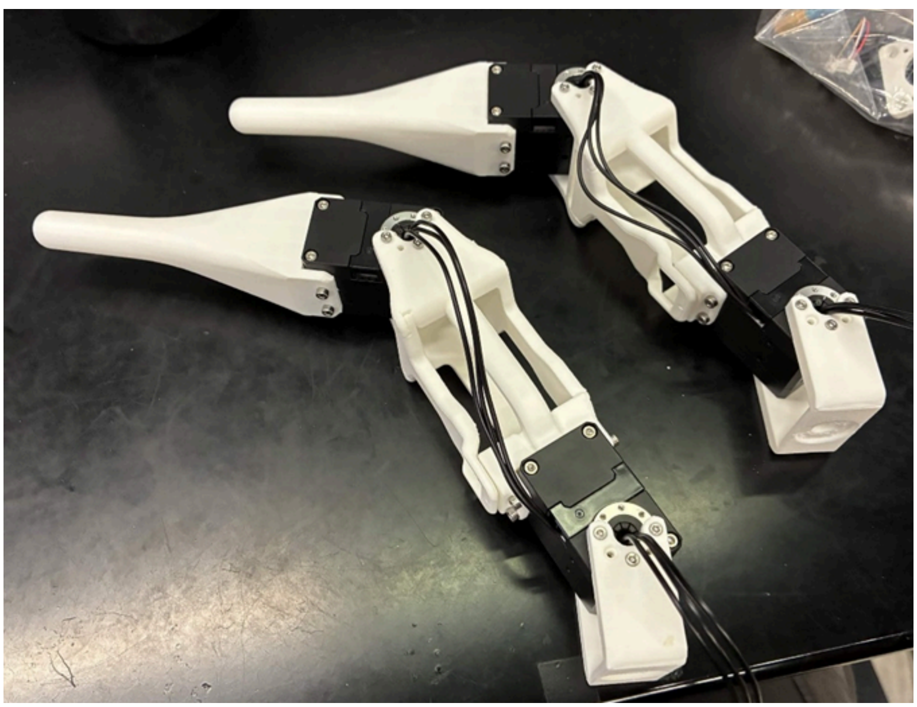

Figure 8: The two assembled Ringbot legs

Attaching one of these legs to the primary drive module yields a fully completed leg.

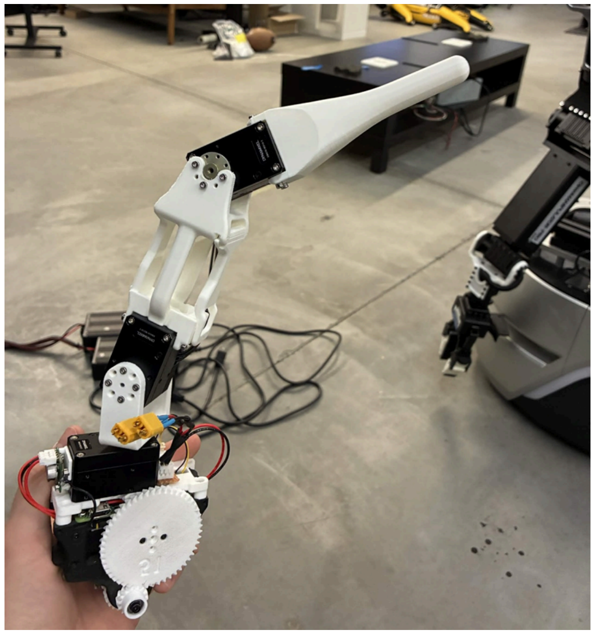

Figure 9: The two assembled Ringbot legs

5. Discussions

This project involved much trial and error, particularly regarding additive manufacturing settings and electromechanical integration. Dialing in the correct slicing settings for TPU on the Raise3D E2 proved critical, specifically discovering that lowering the dense support fill and enabling cooling fans ultimately solved the separation issues. Additionally, physical space constraints demanded iterative CAD adjustments. Soldering the perfboard also highlighted the volatility of using simple lines of solder, proving that stripped solid core wire was a far more reliable method for connecting power and data ports. These iterative fixes were vital to achieving smooth mechanical operation and reliable electrical connections.

6. Future Work

After attaching the leg to the primary module, one leg of the Ringbot was officially completed. This marks the final milestone I was able to complete this semester, I unfortunately ran out of time to finish the secondary drive module. That being said, the future work for this project consists of first finishing the secondary drive module, which is already partially completed. The IMU and telephone cable connections need to be attached, then the 3D printed Raspberry Pi stand-in can be attached. The U2D2 can then be attached to the body cover. Another perfboard needs to be made and attached, and another battery connector needs to be made and attached as well. Finally, all data, power, and ground connections need to be connected to the perfboard, and the second module will be completed. At this point, the project will be ready to move on to testing.

7. References

- Daniel Feutz “Ringbot Circuit Schematic”
- Daniel Feutz “Ringbot Fabrication Document v2”
- J. Kim et al., "Ringbot: A Monocycle Robot with Legs," *IEEE Robotics and Automation Letters*, 2024. Available: https://ieeexplore.ieee.org/document/10423226